

Euler's Formula: $r = e - v + 2$ (must be connected or else the infinite region counted twice)

i) $e=0$ so $r=1$
 $v=1$

ii) $e=1$ so $r=1$ if $e=2$, $r=2$ so Euler's formula applies for Both simple and multigraphs.

iii) $e=3$ so $r=2$

k_3 $v=3$

iv) $e=4$ so $r=2$
 $v=4$



v)



SINCE

$v=5$

$e=10 = \frac{4 \cdot 5}{2}$ (Handshaking lemma)

$r=7$

Since a ~~non~~ planar graph has an edge that intersects another edge, Euler's formula doesn't apply. This edge can create an infinite number of fake regions by adding fake vertices as well. "False vertices" are intersections between 2 edges caused by a non-planar graph where a vertex is not there. ^{both simple & multi.}

∴ Euler's Formula applies for planar, connected graphs.

~~A~~ ^{dual} ~~graph~~ for a (planar) map is only simply connected. So Euler's formula applies for regions on a plane, a map.

4CT: Any planar, simply connected graph can be coloured using less than or equal to 4 colours.

consider a maximally planar graph as the counterexample. A maximally planar graph is a ^{simple, planar} graph where adding 1 edge violates planarity.

- AKA triangulation graphs.

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 C Feb 24

Disprove: For every set of lines in a plane, where no two of the lines in the set are parallel, there exists a common intersection point. Use Induction.

$\forall n \geq 2, P(n)$, where $P(n)$ is the proposition that every n lines in a set on a plane, no two of which are parallel, have a common point, for $n \geq 2$.

1. $\forall n \geq 2, P(n)$

2. Basis step. For $n=2$ common point



3. Inductive step. Assume $P(k)$ is true for an arbitrary fixed integer $k \geq 2$. I.H. That a set of k lines, of which no two are parallel, share a common point for k lines by I.H.

4) Consider $k+1$ lines where $P(k)$ is true[^] and the added line is translated from the previous common point of ~~$P(k)$~~ , Hence, the set of $k+1$ lines does not satisfy $P(k+1)$.

5) This completes the inductive step.

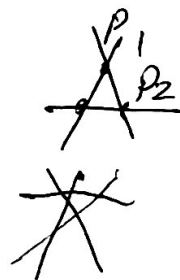
6) By disproof of math. ind., $P(n)$ is not true for all integers $n \geq 2$.

- WRONG Disproof.

disproof is in: the point of intersection between $P(k) \neq P(k+1)$, but can not be a line for distinct n lines,

So $P_2 \neq P_1$, but for $k=2$ this breaks

$\therefore$ ind. step shows $k \geq 3$, not $k \geq 2$



every region has ≥ 3 edges on a simple connected, planar graph with ≥ 3 vertices.

$$\Rightarrow 2e \geq 3r$$

every edge is counted twice for a planar graph (simple) when considering regions
 ~~$\Rightarrow 2e \geq 3r$~~
 ~~$\sum_{i=1}^n \deg(r_i) \geq 3r$~~ or $2e = \sum_{i=1}^n \deg(r_i) (= \sum_{i=1}^n \deg(v_i))$ - every edge has 2 adj. regions, - both count edges twice.

- assume every region has 3 edges $2e = 3r$, for a maximally planar graph then

$$\Rightarrow 2e = 3e - 3v + 6, \text{ Euler applies } \therefore \text{ simple, planar graph}$$

$$\Rightarrow e \leq 3v - 6$$

max

max

Counterexample for 4CT should be maximally planar so the most number of unique colours is used.

$$\therefore e = 3v - 6 \text{ for our maximally planar graph}$$

$$\sum_{i=1}^n \deg(v_i) = 2e = 6v - 12$$

examples



$$\frac{\sum_{i=1}^n \deg(v_i)}{v} = 6 - \frac{12}{v}$$

average number of adjacent vertices per vertex in a maximally planar graph

~~our counterexample should have 1, 2, 3, 4, 5 adjacent edges, - for $\deg(v) \leq 5$, 4CT is true.~~

$\therefore$ every maximally planar graph has a vertex with less than 6 neighbours.

Strong Induction: $P(n)$ is true $\forall n > 0$

base step: $P(1)$ is true

inductive step: $[P(1) \wedge P(2) \wedge \dots \wedge P(k)] \Rightarrow P(k+1)$ is true $\forall k > 0$.

ex 42, 28

use strong induction to show that a simple polygon with n sides, $n \geq 3$, can be triangulated into $n-2$ triangles.

n edges, n vertices, $r = e - v + 2$ (r = # of regions) $\rightarrow 2v = 3 + e$
 $e - v + 1 = k - 2$ $e = \frac{3}{2} + \frac{e}{2}$

if Theorem is true, to triangulate a polygon requires $(e-v) = v-3$

Theorem says that for a polygon w/ n vertices, edges, $n-3$ edges are added to triangulate a polygon.



well ordering, induction & strong induction are equivalent.

~~conjecture:~~

A maximal planar graph is the same as a triangulated polygon.

A polygon with n sides can be triangulated into $n-2$ triangles

$\therefore n-1$ regions in total

$$r = e - v + 2$$

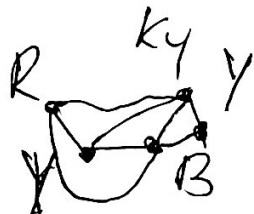
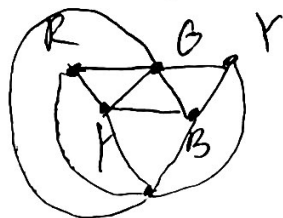
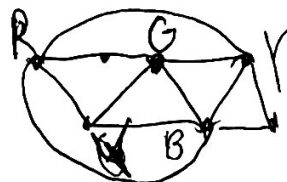
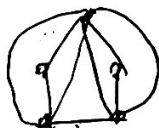
$$n-1 = e - v + 2$$

$$v-1 = e - v + 2$$

$$2v = e + 3$$

$$e = 2v - 3 \neq 3v - 6$$

Disproved.



Prove 4CT by strong Induction:

let $P(n)$ be the proposition that 4CT is true for all positive integers $n \geq 3$, where n is the number of vertices of a planar, simply connected graph.

~~lemma~~ lemma: If a graph G is maximally planar and has n vertices is 4-colourable, then a graph G_1 with n vertices is 4-colourable, $e = 3n - 6$ for $n \geq 3$

basis step: $P(3)$



inductive step: IH: $[P(3) \wedge P(4) \wedge \dots \wedge P(n-1)]$ is true.

Assuming IH and adding a vertex to a graph with $n-1$ vertices to form a graph with n vertices. Then make Graph G_n maximally planar. G_n should be uncoloured.

Since there exists a vertex with ≤ 6 neighbours, using Kempe's algorithm, that ~~node~~ ^{vertex} can be made 4 colourable with its neighbours. Then removing that vertex and reforming a graph with $n-1$ vertices has to be 4 colourable by IH.

- use strong induction since $\exists$ a vertex with $\deg(v) \leq 6$ so the removal $\rightarrow$ maximal planarity can be applied $n-1$ times, ~~satisfying~~

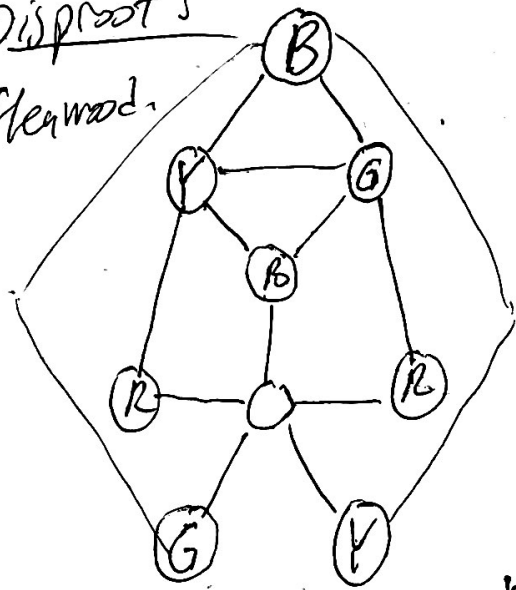
IH,

By basis & inductive step, $P(n)$ is true for all $n \geq 3$. This completes the proof by mathematical induction.

$n=1, 2$ are 4-colourable

This is Kempe's proof.

Disproof's
Heuristics



Kern's recursive Algorithm to 4 color a graph.

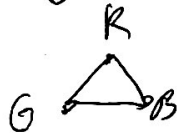
- 1) pick a vertex that is uncoloured and has < 6 neighbours (maximally planar)
- 2) use below to colour
- 3) remove the vertex and make G maximally planar and its edges
- 4) skip 1

worst cases are considered below:

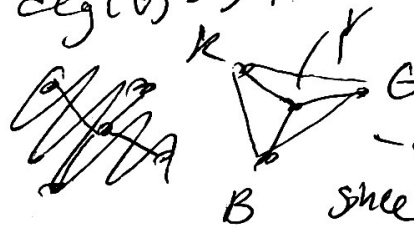
$\deg(v) = 1$:



$\deg(v) = 2$:



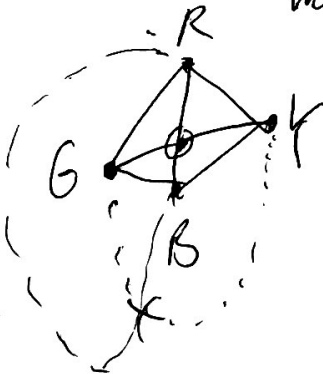
$\deg(v) = 3$:



- must have ring
since our graph is a
dual graph, maximal
planarity helps.

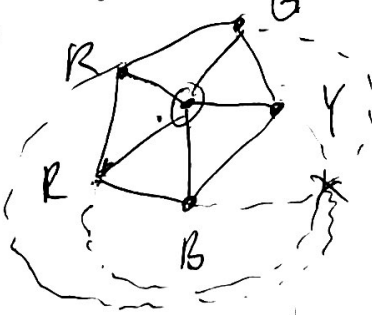
$\deg(v) = 4$:

worst case

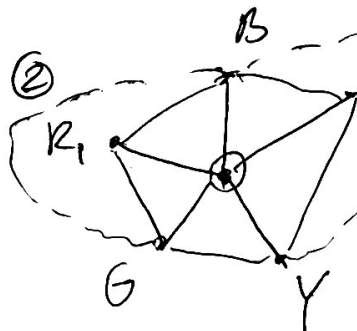


if kern chain, toggle other, which has not kern chain
- kern chain is a wall

$\deg(v) = 5$:



1. toggle B & G/Y



2) b, otherwise, $R_2 = G$, $R_1 = Y$,
by kern chain (4.11).

How does this prove

5CT? $\deg(v) = 4$
considered again